Part 2: Bayesian Probability Analysis

IMDb Movie Reviews Dataset — Formative 3

1. Keyword Selection

We selected three keywords believed to indicate positive sentiment and three believed to indicate negative sentiment, based on how strongly and consistently each word appears in one sentiment class over the other.

Positive indicators: excellent, wonderful, brilliant

Negative indicators: awful, horrible, boring

2. Choice of Conditional Probability

Our group calculates **P(Positive | keyword)** for every keyword, positive-indicating and negative-indicating alike. We do not compute P(Negative | keyword) anywhere in this analysis. For negative-indicating keywords, a *low* P(Positive | keyword) value itself serves as evidence of negative sentiment — no second calculation is needed.

3. Bayes' Theorem

$$P(\text{Positive} \mid \text{keyword}) = P(\text{keyword} \mid \text{Positive}) \times P(\text{Positive}) / P(\text{keyword})$$

Term	Name	Meaning
P(Positive | keyword)	Posterior	What we solve for
P(keyword | Positive)	Likelihood	How common the word is in positive reviews
P(Positive)	Prior	Baseline probability of a positive review
P(keyword)	Marginal	How common the word is overall

4. Dataset Statistics

Total Reviews: 50,000 | Positive: 25,000 (50%) | Negative: 25,000 (50%) | Prior P(Positive) = 0.5000

5. Computation and Presentation

For each keyword, the four probabilities are shown below, followed by the Bayes' Theorem calculation and interpretation.

Keyword: 'excellent'

Probability	Formula	Value
Prior	P(Positive)	0.5000
Likelihood	P(keyword | Positive)	0.11472
Marginal	P(keyword)	0.07104
Posterior	P(Positive | keyword)	0.8074

Calculation: P(Positive | 'excellent') = (0.11472 × 0.5000) / 0.07104 = **0.8074**

Interpretation: A review containing 'excellent' has a **80.7% probability of being positive** (found in 2,868 positive reviews vs. 684 negative reviews).

Keyword: 'wonderful'

Probability	Formula	Value
Prior	P(Positive)	0.5000
Likelihood	P(keyword | Positive)	0.09032
Marginal	P(keyword)	0.05560
Posterior	P(Positive | keyword)	0.8122

Calculation: P(Positive | 'wonderful') = (0.09032 × 0.5000) / 0.05560 = **0.8122**

Interpretation: A review containing 'wonderful' has a **81.2% probability of being positive** (found in 2,258 positive reviews vs. 522 negative reviews).

Keyword: 'brilliant'

Probability	Formula	Value
Prior	P(Positive)	0.5000
Likelihood	P(keyword | Positive)	0.06348
Marginal	P(keyword)	0.04176
Posterior	P(Positive | keyword)	0.7601

Calculation: P(Positive | 'brilliant') = (0.06348 × 0.5000) / 0.04176 = **0.7601**

Interpretation: A review containing 'brilliant' has a **76.0% probability of being positive** (found in 1,587 positive reviews vs. 501 negative reviews).

Keyword: 'awful'

Probability	Formula	Value
Prior	P(Positive)	0.5000
Likelihood	P(keyword | Positive)	0.01136

Marginal	P(keyword)	0.05766
Posterior	P(Positive | keyword)	0.0985

Calculation: P(Positive | 'awful') = (0.01136 × 0.5000) / 0.05766 = **0.0985**

Interpretation: A review containing 'awful' has only a 9.8% probability of being positive, i.e. **90.1% likely negative** (found in 2,599 negative reviews vs. 284 positive reviews).

Keyword: 'horrible'

Probability	Formula	Value
Prior	P(Positive)	0.5000
Likelihood	P(keyword | Positive)	0.01304
Marginal	P(keyword)	0.04202
Posterior	P(Positive | keyword)	0.1552

Calculation: P(Positive | 'horrible') = (0.01304 × 0.5000) / 0.04202 = **0.1552**

Interpretation: A review containing 'horrible' has only a 15.5% probability of being positive, i.e. **84.5% likely negative** (found in 1,775 negative reviews vs. 326 positive reviews).

Keyword: 'boring'

Probability	Formula	Value
Prior	P(Positive)	0.5000
Likelihood	P(keyword | Positive)	0.02364
Marginal	P(keyword)	0.06102
Posterior	P(Positive | keyword)	0.1937

Calculation: P(Positive | 'boring') = (0.02364 × 0.5000) / 0.06102 = **0.1937**

Interpretation: A review containing 'boring' has only a 19.4% probability of being positive, i.e. **80.6% likely negative** (found in 2,460 negative reviews vs. 591 positive reviews).

6. Implementation Notes

Bayes' Theorem was implemented in pure Python using only the built-in **csv** and **re** modules — no external machine learning or data-science libraries (e.g. no pandas, numpy, or scikit-learn) were used for the probability calculations. Keyword matching uses whole-word regular expressions (\b boundaries) so that, for example, 'excellent' does not incorrectly match inside 'excellently'. Each review is checked for keyword *presence* (not frequency), consistent with a standard bag-of-words presence model.

7. Keyword Justification

All three positive keywords produced posteriors of 0.76–0.81, and all three negative keywords produced posteriors of 0.10–0.19. This symmetric, strong separation from the 0.50 baseline in both directions confirms these are genuine sentiment-indicating words rather than noise terms that would sit close to 0.50.